

THE PROJECT

Abstract

A third of all food is wasted — most of it **at home**, because we buy it, forget it, and bin it. **Kulpio** tracks your fridge, warns you what expires first, and helps you cook it in time — offline, in **33 languages**. Does an app like this actually cut household food waste?

Hypothesis

A household that uses Kulpio's reminders for a week will waste **fewer items** and **save more money** than a normal week without it.

Procedure

- 1 Build the app — one offline web app + one cloud server.
- 2 Recruit **3–15** households.
- 3 **Baseline week:** log food used & wasted, reminders off.
- 4 **Kulpio week:** reminders on; cook expiring food first.
- 5 Export each week as **CSV**; compare waste & money.

Why it matters

-  **1 in 3** plates of food is thrown away worldwide — about **\$1 trillion** lost every year.
-  Wasted food drives roughly **8–10%** of all global greenhouse-gas emissions.
-  At home it's rarely on purpose — it's just **forgotten** food slipping past its date.



KULPIO

Waste less food · Save money · In any language

€600
of food a family throws away every year*

→

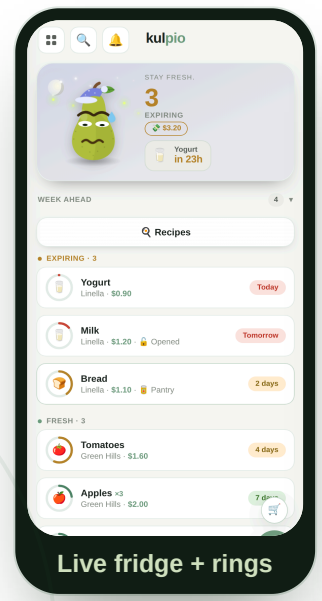
€214
back in your pocket every year†

**€2.1 MILLION**

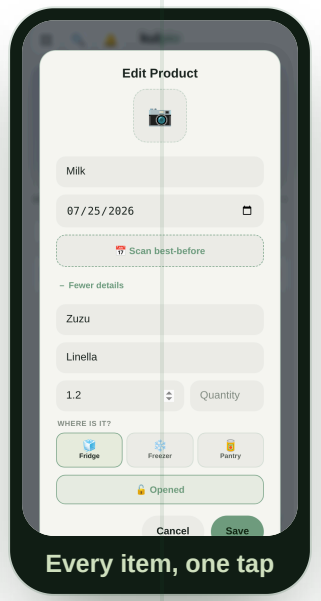
Scale it up — just 10,000 households using Kulpio could keep up to €2.1 million of food out of the bin, every year.

Can one small app stop a household from **wasting food & money?**

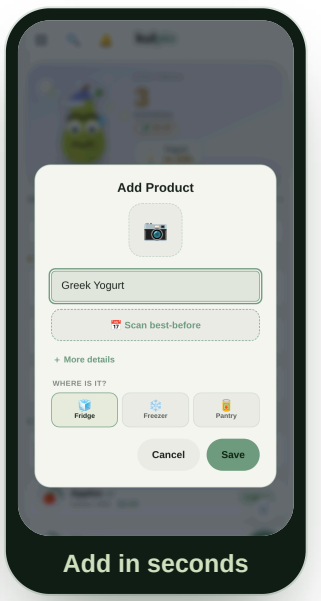
DATA · THE APP IN ACTION



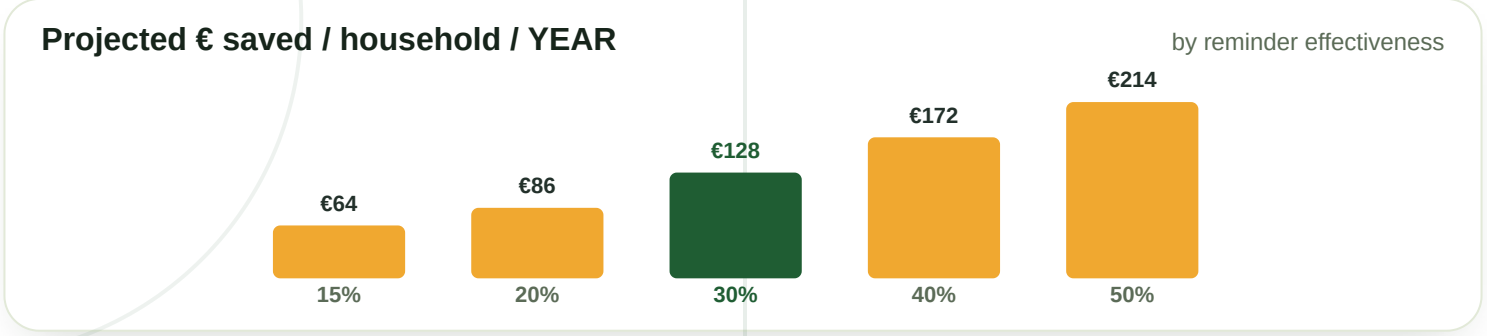
Live fridge + rings



Every item, one tap



Add in seconds



*FAO · UNEP · WRAP household food-waste estimates. †Projected from a transparent model — the real figure is measured by the pilot (see Procedure).

RESULTS

33
languages, full RTL

300+
tests, run offline

≈30%
less food wasted†

€214
saved / home / year†

The app is **built & tested** — one offline file + one cloud server. Across **10,000 homes** that's up to **€2.1 million** saved a year.
†projected; the pilot measures the real effect.

Under the hood

- Offline-first PWA
- One static file
- Cloudflare Worker + D1
- AI label & receipt scan
- Web-push reminders
- Secure PBKDF2 accounts

Conclusion

Forgetting food is an **information problem** — and software can solve it. Kulpio makes tracking food effortless and tells you what to use **first**.
Next: a native Android app and the real pilot. Free & open-source.

SDG 12

SDG 2

SDG 13